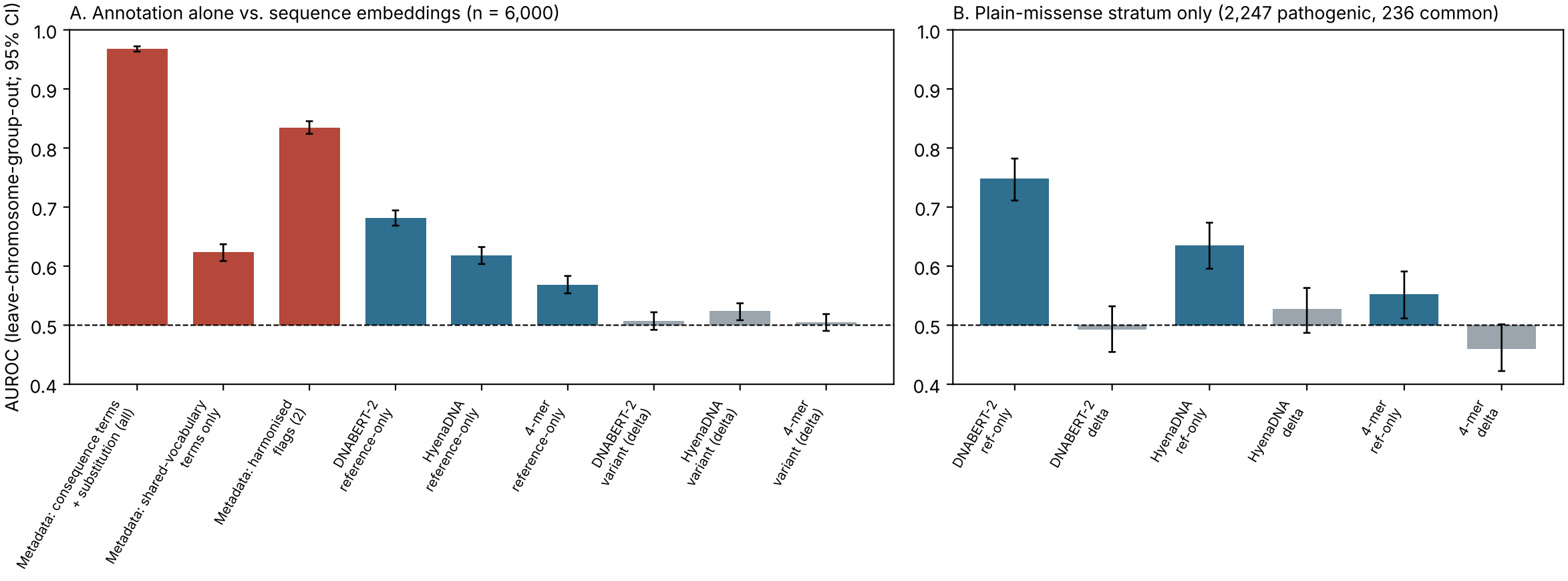
A. Annotation alone vs. sequence embeddings (n = 6,000)
B. Plain-missense stratum only (2,247 pathogenic, 236 common)
AUROC (leave-chromosome-group-out; 95% CI)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Metadata: consequence terms + substitution (all)
Metadata: shared-vocabulary terms only
Metadata: harmonised flags (2)
DNABERT-2 reference-only
HyenaDNA reference-only
4-mer reference-only
DNABERT-2 variant (delta)
HyenaDNA variant (delta)
4-mer variant (delta)
DNABERT-2 ref-only
DNABERT-2 delta
HyenaDNA ref-only
HyenaDNA delta
4-mer ref-only
4-mer delta